

MIN-MAX ± 1 QUADRATIC FORM

Update · MO 413935 · existence of the limit still OPEN

THE QUESTION (unchanged)

$$L = \lim_{n \rightarrow \infty} n^{-3/2} \min_{a_{ij} = \pm 1} \max_{x_i = \pm 1} \left| \sum_{i < j} a_{ij} x_i x_j \right|$$

$\alpha_n = m_n/n^{3/2}$. Sandwich proved; existence of $L = \lim \alpha_n$ still open.

PROVED — SANDWICH (load-bearing)

$$\frac{1}{\pi} \leq \liminf \alpha_n \leq \limsup \alpha_n \leq \frac{1}{2}$$

Lower: dual-Gaussian arcsine (every Seidel matrix).

Upper: denseness + Paley / conference. Does not force $\liminf = \limsup$.

PATH TO $L = 1/2$ (if it exists this way)

On dense $\rho = 1$ Paley family $n = p^2 + 1$: $m_n \geq \Phi - 2 \Rightarrow E(1) \Rightarrow \text{denseness} \Rightarrow L = \frac{1}{2}$.

$\rho = 1$ is proved (halfspace boolean evec $Cx = px$). $E(1)$ is the blocker — reduced to residual structure on Max

WHERE WE ARE (honest)

- Sandwich + $\rho = 1$ family + denseness: shipped.
- Bi-tight / Type-I covers: reduced to $g_{\min} \geq L(p)$ or $\lambda_2(P \odot P) \leq 4/N$.
- Certified $g_{\min} \geq L(p)$ at $p = 5, 7$ ($n = 26, 50$). General $p \geq 5$: OPEN.
- Deep non-tight ND / $\Phi \geq \Phi(C)$ for $p \geq 5$: still OPEN.
- Main Theorem $L = \frac{1}{2}$: not claimed. Soft-close banned.

RESIDUAL ATTACK · WHAT'S NEW

Max+ fourth moments · operator T · bi-tight residual · still not a prize solution

NEW STRUCTURE (Props 15.47-15.69, sketch)

- ▶ $g_{\min} = -\max|m_4|$ on 4-sets with $|\kappa| = 1$ (pairing algebra).
- ▶ Master identity: $m_4 = \kappa/p^2 + \text{Ext}/(4p)$, $\text{Ext} = Tm_4$.
- ▶ Combinatorial: $T\kappa = 0$ on every $|\kappa| = 1$ set (conference $C^2 + K_4$).
- ▶ Spectrum: $\lambda_{\max}(T) = 4p$ at $p = 5, 7 \Rightarrow 4pI - T$ singular;
- ▶ $m_4 = m_* + h$ with $h \in \ker(4pI - T)$; min-norm m_* already $\leq L$ there.
- ▶ Equivalent forms: $Q \leq 16N\|B\|_F^2 \Leftrightarrow \lambda_{\text{cycle}} \leq 8 \Leftrightarrow \lambda_2(P \odot P) \leq 4/N$.

CERTIFIED NUMBERS (multi-worker)

$p = 5$: $g_{\min} = -3/65 \geq L(5) = -3/50 > T(5)$. $p = 7$: $g_{\min} = -109/2863 \geq L(7) = -5/98$.

$p = 3$ equality case for $16N / \lambda_{\text{cycle}} = 8$; gap holds numerically $p = 5, 7$ (fails $p = 3$).

STILL OPEN (blockers)

- ▶ P0 Prove $g_{\min} \geq L(p)$ (or $\lambda_2(P \odot P) \leq 4/N$) for all primes $p \geq 5$.
- ▶ P1 Deep non-tight ND: $\Phi \geq \Phi(C)$ outside Max+-tight covers.
- ▶ P2 Only if P0+P1: Main Theorem $L = \frac{1}{2}$ (or a genuine non-existence pair).

STATUS

Existence of L remains OPEN. Structure is tighter; the last mile is still proof.